



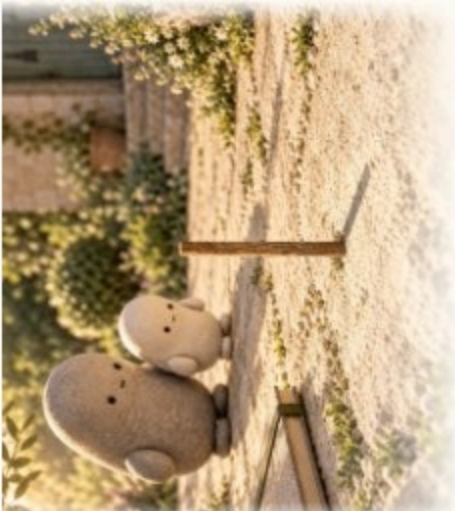
Little Wonderer found a long shape beside a standing stick. It was a shadow: a place where the stick blocked sunlight, the light from the sun. "Will it stay this long?" Little asked. Big set a linen notebook beside the garden path. They had a question to keep.



Little took a piece of chalk, a soft stick for making marks. "First mark the tip," said Big. Little drew a small line where the shadow ended. To measure is to find how long something is. They counted small, even footsteps from the wooden stick to the chalk.



At noon, the middle of the day, they came back. The wooden stick had stayed in place, but its shadow had changed. "It is short and sharp!" said Little. Around noon, the sun is near its highest point for the day. The morning shadow had reached much farther.



"Which way does it point?" asked Big. Little studied the ground. Sunlight reached one side of the stick. The shadow stretched away on the other side. The bright side faced the sun; the shadow pointed away from the light. Little traced its edge without looking up at the sun.

Shadows at Noon

The shadow that changed



A short shadow. A high sun.

GRADE 3

Make a little book.

1. Print one-sided: US Letter, portrait, actual size / 100%. No fit. Headers/footers off.
2. Fold long edge to long edge; reopen. Fold short edge to short edge, then in half again. Open: eight boxes.
3. Cut only the solid center line across the middle two boxes. Keep the outer boxes joined.
4. Refold long edge to long edge, print outside. Push ends together to open a diamond, then a cross.
5. Wrap into a book with page 1 in front. Follow pages 1 to 8.



Measure, notice, and pause whenever you like.

Your turn

Write one command sentence about measuring a shadow.

Example: First stand the stick, then mark the tip of the shadow.



Little marked the new tip with chalk. This time, they used a measuring stick to check both lengths. They began at the same standing stick and used the same unit each time. The noon length was shorter. "The stick stayed the same," Little said. "The sun's height changed."